

1. SYSTEM OVERVIEW & PROBLEM STATEMENT CONTEXT

Uniformed personnel serving in the **Central Armed Police Forces (CRPF, BSF, CISF, ITBP, SSB)**, the **Indian Armed Forces**, and **State Police formations** face prolonged operational conditions—hazardous terrain, remote postings, circadian disruptions, and intense operational tempo. Conventional stress identification across security forces remains fundamentally reactive, relying on lagging peer observation after critical psychological burnout or disciplinary incidents manifest.

SentinelWell AI transforms this paradigm from **reactive intervention to proactive prevention**. By integrating non-invasive, objective administrative duty records (consecutive duty days, night shift cycles, deployment duration, leave denial patterns) with voluntary self-reported wellness telemetry, the platform computes a continuous 4-band **Welfare Risk Index** backed by **Explainable AI (TreeExplainer SHAP)**. It empowers welfare officers to initiate timely rest cycles and counseling while strictly safeguarding personnel confidentiality and ensuring zero punitive surveillance.

2. DISTRIBUTED SYSTEM TOPOLOGY & ARCHITECTURE

The system utilizes a 3-tier decoupled microservices topology engineered for high resilience, zero-leakage RBAC boundaries, and sub-15ms ML inference:



Figure 1A: Comprehensive System Topology & Microservices Component Architecture.

2.1 END-TO-END DATA & INFERENCE FLOW

The sequence below models the operational path from voluntary self check-in submission to real-time ML risk scoring and welfare officer triage:

Step	Origin / Actor	Target / Interface	Operational Action & Payload
01	Personnel (Ground Officer)	Frontend Web Portal	Submits voluntary daily check-in (Stress: 1–10, Sleep: 1–5, Energy: 1–5, Workload: 1–5).
02	Frontend Web Portal	Express API Server	POST /api/wellness/assessment with JWT authorization token.
03	Express API Server	Aiven Cloud MySQL	Persists WellnessAssessment record and queries 30-day duty history.
04	Express API Server	FastAPI ML Service	Dispatches feature vector: [dutyDays, nightShifts, deployDays, leaveRej, dutyHrs, sleepHrs, fatigue].
05	FastAPI ML Engine	Internal Feature Pipeline	Computes derived fatigue variables: SDI (Sleep Deprivation Index) & Burnout Velocity .
06	FastAPI ML Engine	Dual-Model Inference	Executes Isolation Forest for anomaly scoring and LightGBM for multi-class risk probabilities.
07	FastAPI ML Engine	SHAP TreeExplainer	Extracts exact Shapley attributions $ \phi_i $ to identify top 3 primary trigger stressors.
08	FastAPI ML Engine	Express API Server	Returns continuous 0–100 score, risk category ('HIGH'), and natural-language primary triggers.
09	Express API Server	Aiven Cloud MySQL	Stores RiskAssessment entry. If score ≥ 61 , automatically creates a live Alert.
10	Express API Server	Frontend Web Portal	Pushes confirmed assessment to user with updated personal 6-month wellness trajectory.
11	Welfare Officer (WO-208)	Welfare Command Center	Detects alert in triage queue; inspects continuous risk gauge (e.g. 64 High) and SHAP factor bars.
12	Welfare Officer (WO-208)	Express API Server	Issues proactive welfare intervention (e.g., 48-hr Rest Cycle); logged into immutable audit register.

Figure 1B: Structured End-to-End Operational Inference and Triage Sequence.

3. MATHEMATICAL FORMULATION & FEATURE SPACE

The prediction engine ingests 8 primary attributes categorized into administrative duty parameters and voluntary wellness telemetry:

Feature	Type	Domain	Operational Definition
x_1 : consecutive_duty_days	Float	$[0, \infty)$	Continuous duty days logged without a mandatory 24-hr rest window.
x_2 : night_shifts_last_30d	Float	$[0, 30]$	Cumulative night watch duties logged between 2200 and 0600 hrs in past 30 days.
x_3 : deployment_duration_days	Float	$[0, \infty)$	Cumulative active days deployed at forward or high-altitude outposts.
x_4 : leave_rejected_count	Float	$[0, \infty)$	Number of formal leave requests unapproved or deferred in past 12 months.
x_5 : transfer_frequency_2y	Float	$[0, \infty)$	Battalion or company reassignment frequency across past 24 months.
x_6 : duty_hours_weekly	Float	$[0, 168]$	Aggregated duty hours on active roster during a 7-day period.
x_7 : avg_sleep_hours	Float	$(0, 24]$	Voluntary, self-reported average sleep duration per 24-hour cycle.
x_8 : self_reported_fatigue	Integer	$[1, 5]$	Subjective exhaustion rating (1 = Fully Rested, 5 = Severe Exhaustion).

Derived Fatigue Variables

To model non-linear physiological strain, two derived variables are calculated prior to model scoring:

Sleep Deprivation Index (SDI): $SDI = x_1 / x_7 = (\text{consecutive_duty_days}) / (\text{avg_sleep_hours})$
Burnout Velocity (BV): $BV = x_6 \times x_8 = (\text{duty_hours_weekly}) \times (\text{self_reported_fatigue})$

4. DUAL-MODEL MACHINE LEARNING ENGINE & SHAP EXPLAINABILITY

The inference engine pairs unsupervised behavioral anomaly detection with supervised multi-class gradient boosting:

4.1 Anomaly Scoring via Isolation Forest

An Isolation Forest model trained on normative duty scheduling profiles evaluates the augmented feature vector. It computes the anomaly score:

$$s_{iso} = -Score_{iso}([x_1, x_2, ..., x_8, SDI, BV]) \in [-0.5, 0.5]$$

4.2 Multi-Class Risk Classification (LightGBM)

The LightGBM classifier evaluates the 11-dimensional vector $X = [x_1, \dots, x_8, SDI, BV, s_{iso}]$ to generate a calibrated probability distribution across 4 welfare risk tiers:

$$P = [p_{Low}, p_{Mod}, p_{High}, p_{Crit}] \text{ where } \sum_{k=0}^3 p_k = 1$$

4.3 Continuous 0–100 Welfare Risk Formulation

To eliminate discrete step-function jumps, a continuous risk index S is formulated via inner product with risk band weights:

$$S = round(\sum_{k=0}^3 p_k \times w_k) \text{ with centroid vector } W = [0, 33.33, 66.67, 100.00]$$

4.4 Explainable AI (XAI) via TreeExplainer SHAP

For every generated score, exact Shapley values are calculated to identify the top 3 contributing stressors:

$$\phi_i(\hat{y}) = \sum_{S \subseteq F \setminus \{i\}} [\frac{|S|!(|F| - |S| - 1)!}{|F|!}] \times [f(S \cup \{i\}) - f(S)]$$

Primary Triggers: Top 3 features ranked by absolute attribution $|\phi_i|$ (e.g., *Deployment Duration, Night Shifts, Duty Hours*)

5. COMPREHENSIVE MODULE SPECIFICATIONS & UI ARCHITECTURE

Module 1: Welfare Officer Command Center & Real-Time Triage

The operational command center presents high-level force distributions paired with an actionable triage roster.

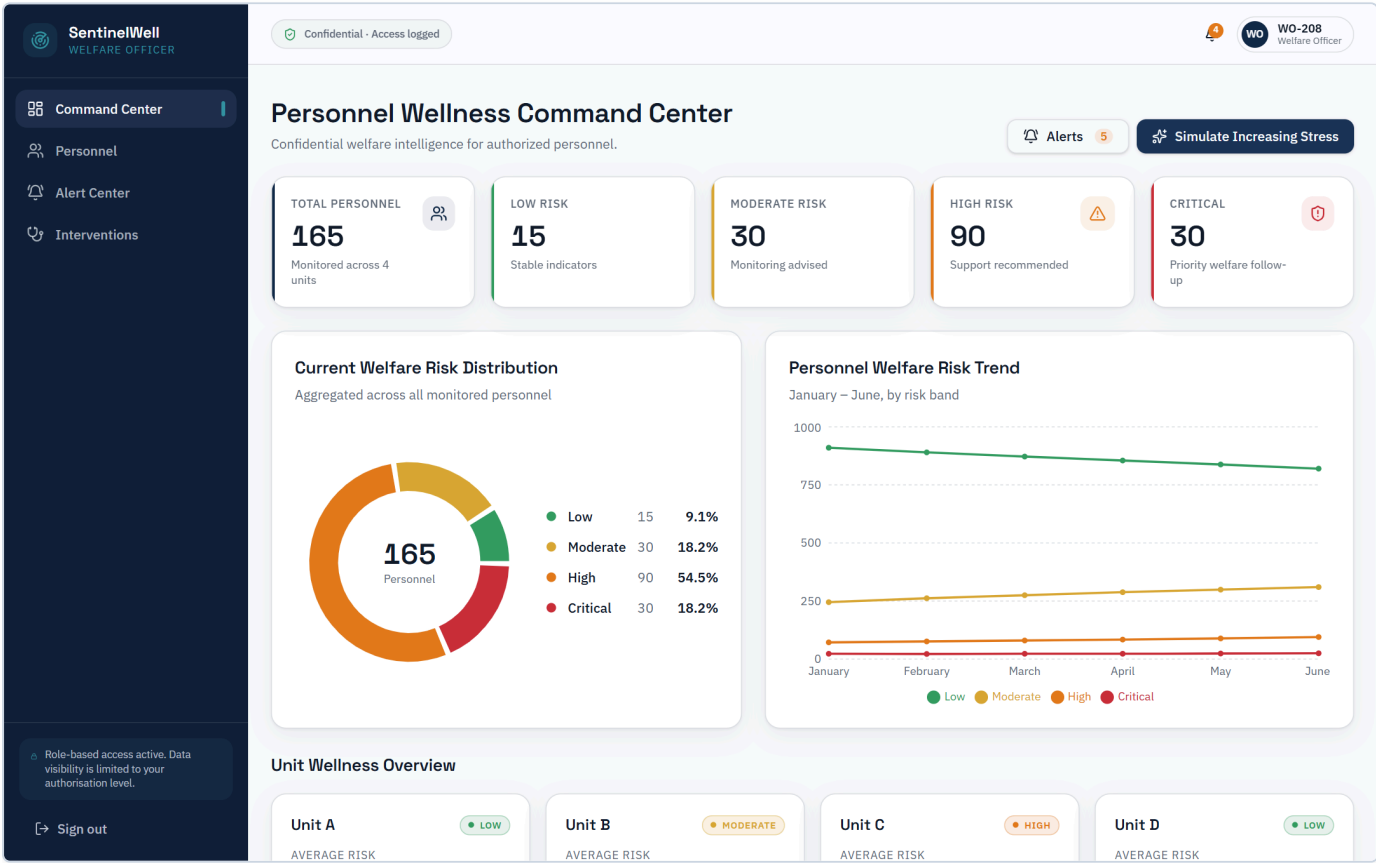


Figure 1: Welfare Officer Command Center displaying macro distributions (165 Monitored, 90 High Risk, 30 Critical) and 6-month historical risk trend.

Below the macro KPI summary, the Command Center displays comparative unit overview cards alongside the High-Risk Personnel Master Table:

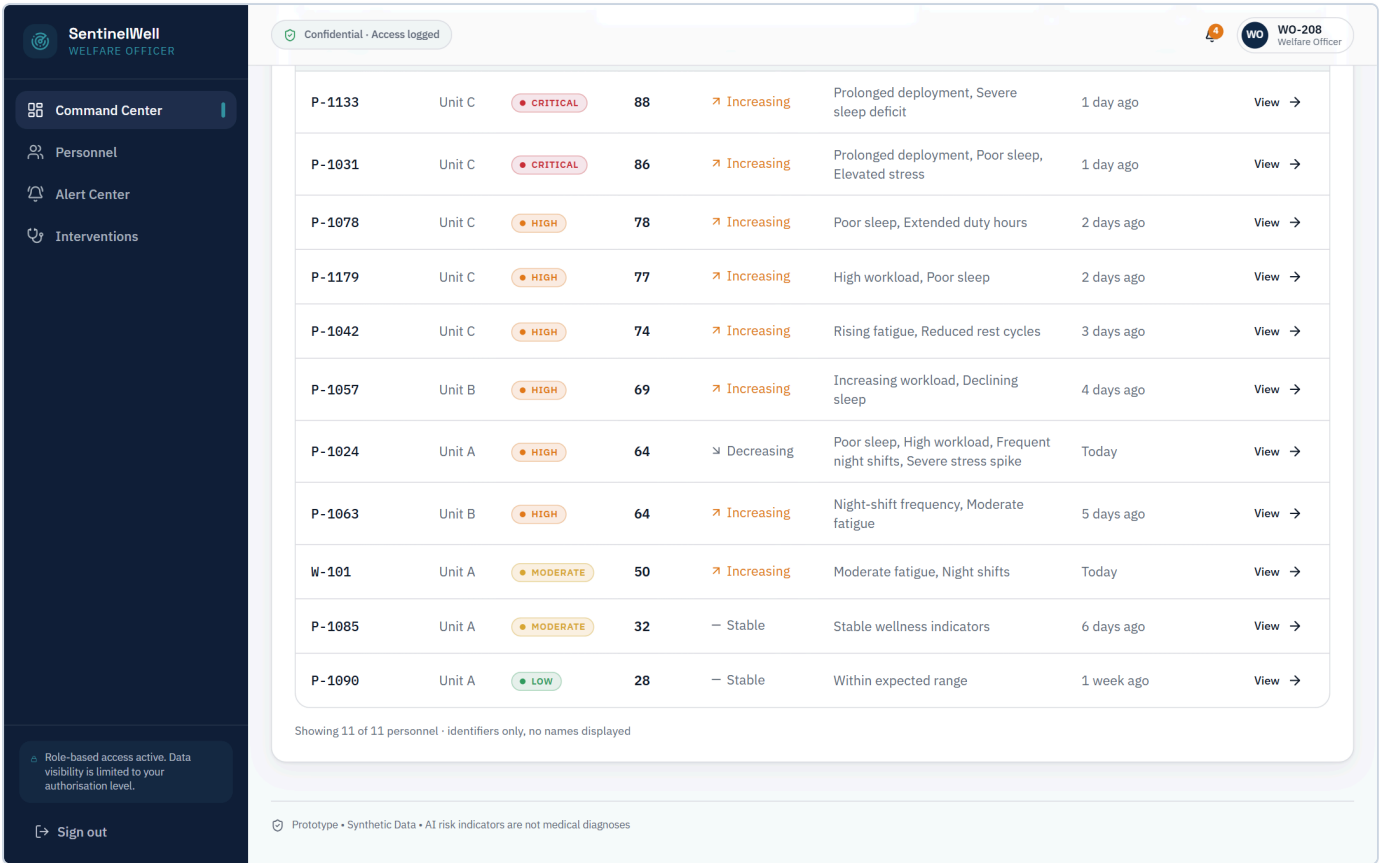


Figure 2: High-Risk Personnel Master Triage Roster displaying Personnel IDs (P-1133, P-1031, P-1078, P-1024), continuous risk scores, trajectories, and direct triage links.

Module 2: Personnel Risk Radar & Explainable AI (XAI)

Provides deep, transparent insight into individual risk drivers without psychiatric labeling. The upper viewport presents the continuous risk gauge and relative SHAP factor attribution bars:

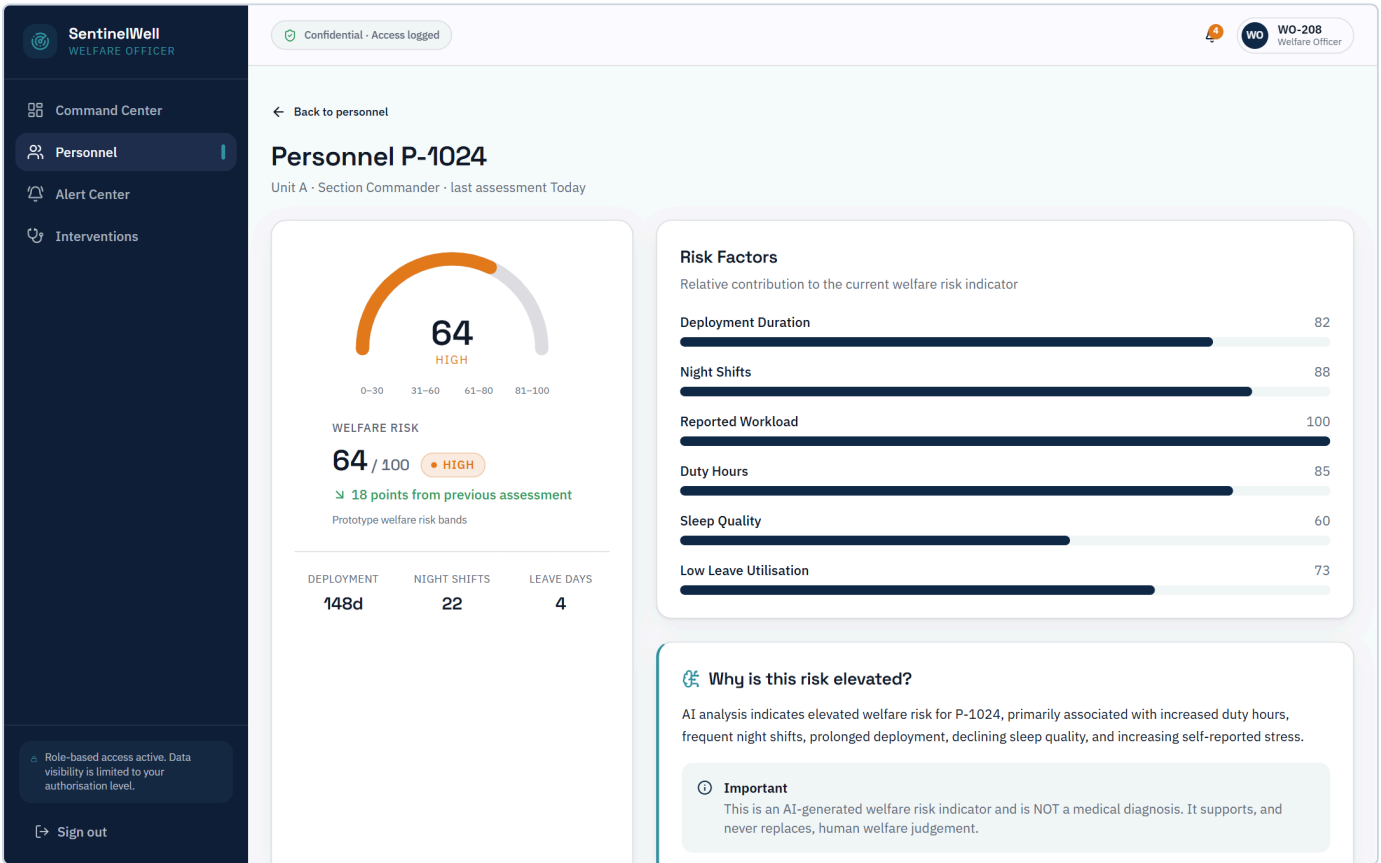


Figure 3: Personnel Risk Radar (P-1024) showing continuous 0–100 gauge (64 High) and normalized SHAP factor attribution (Reported Workload 100%, Night Shifts 88%, Duty Hours 85%).

The lower section of the Risk Radar tracks 6-month longitudinal telemetry trends and provides dedicated human-in-the-loop triage action triggers:

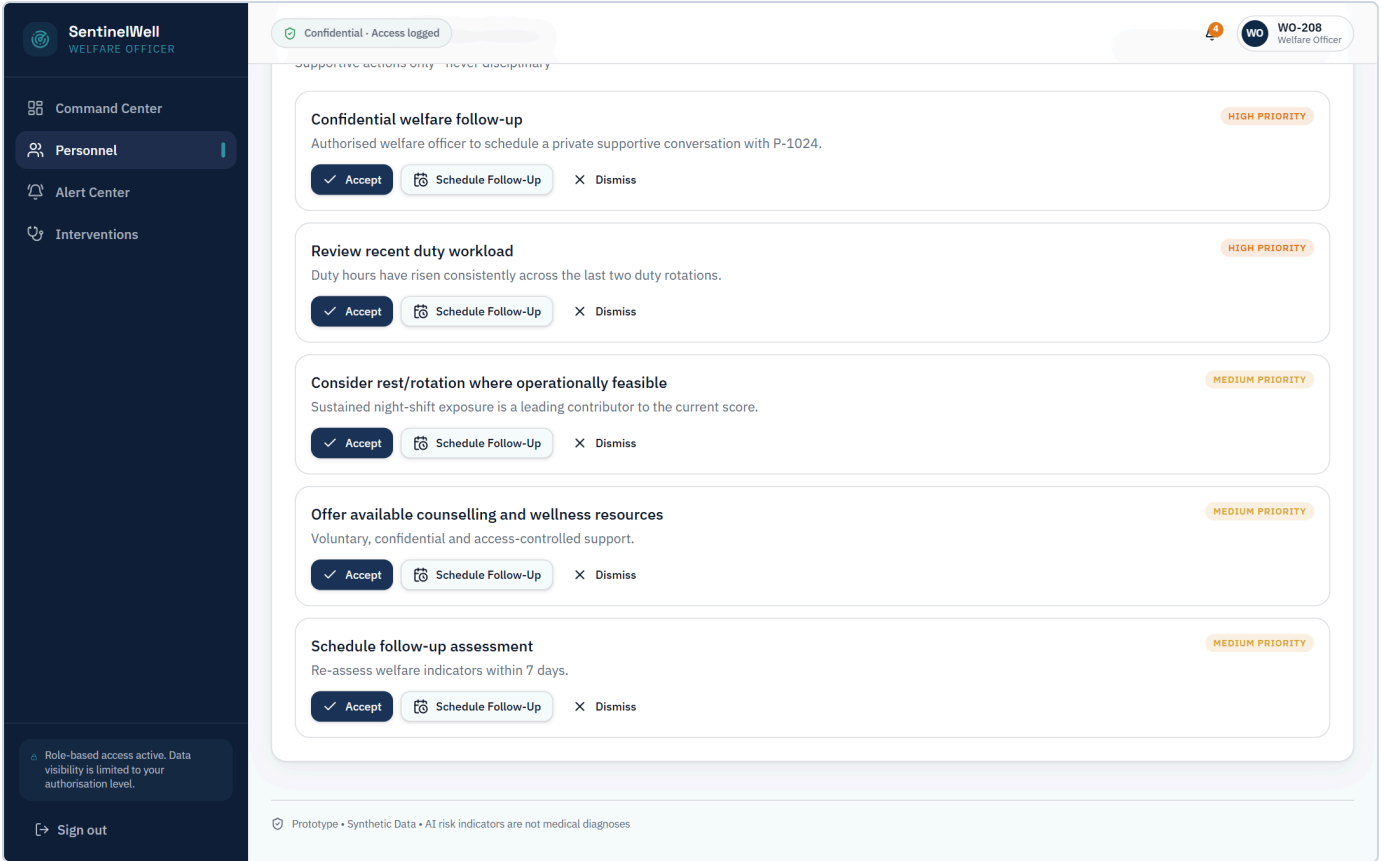


Figure 4: Longitudinal 6-Month Telemetry Trends (Stress, Sleep, Workload, Duty Hours) alongside the Recommended Welfare Actions triage hub.

Module 3: Proactive Welfare Interventions & Live Alert Stream

Enables authorized caseworkers to track intervention lifecycles and monitor automated stress alerts in real time:

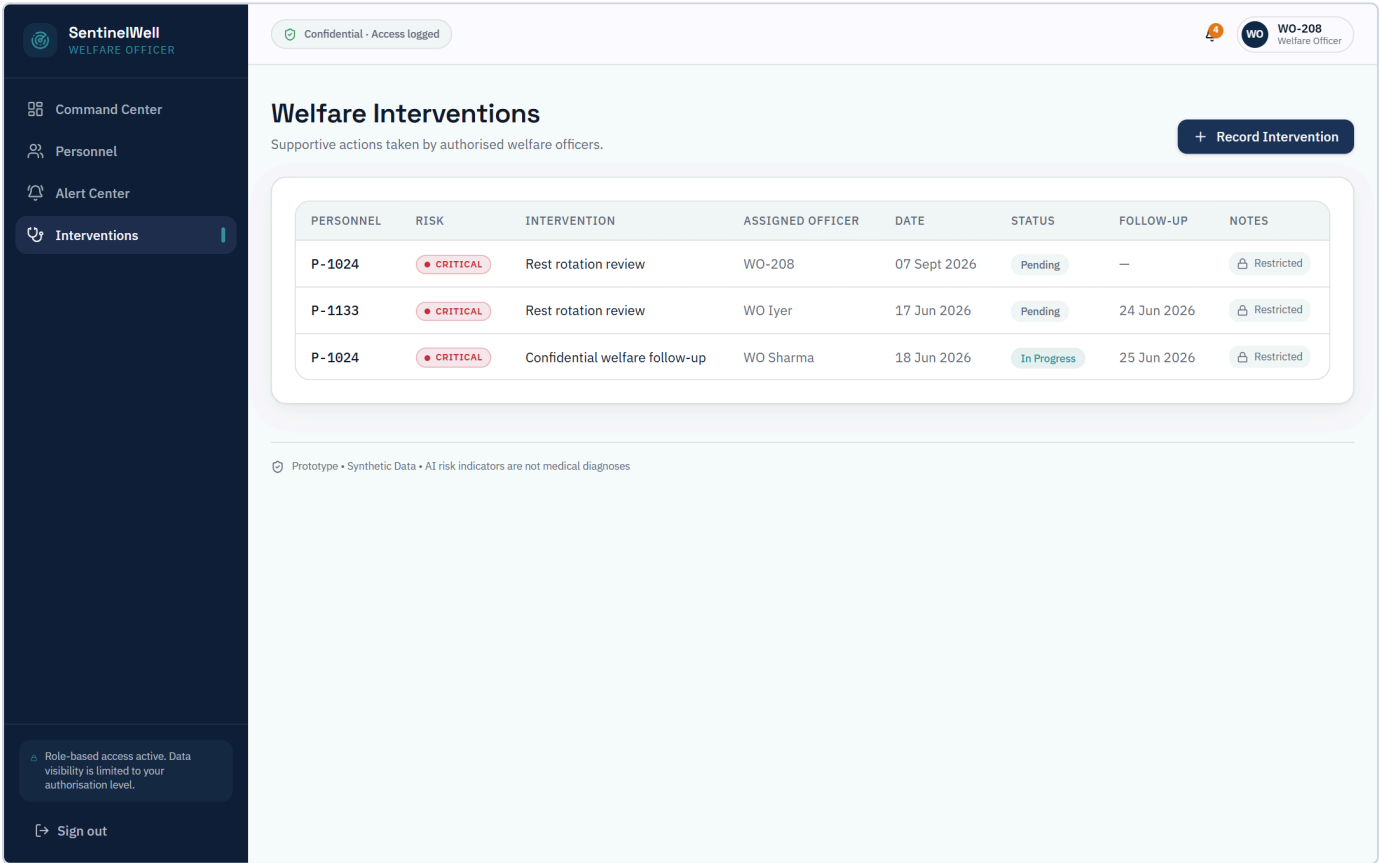
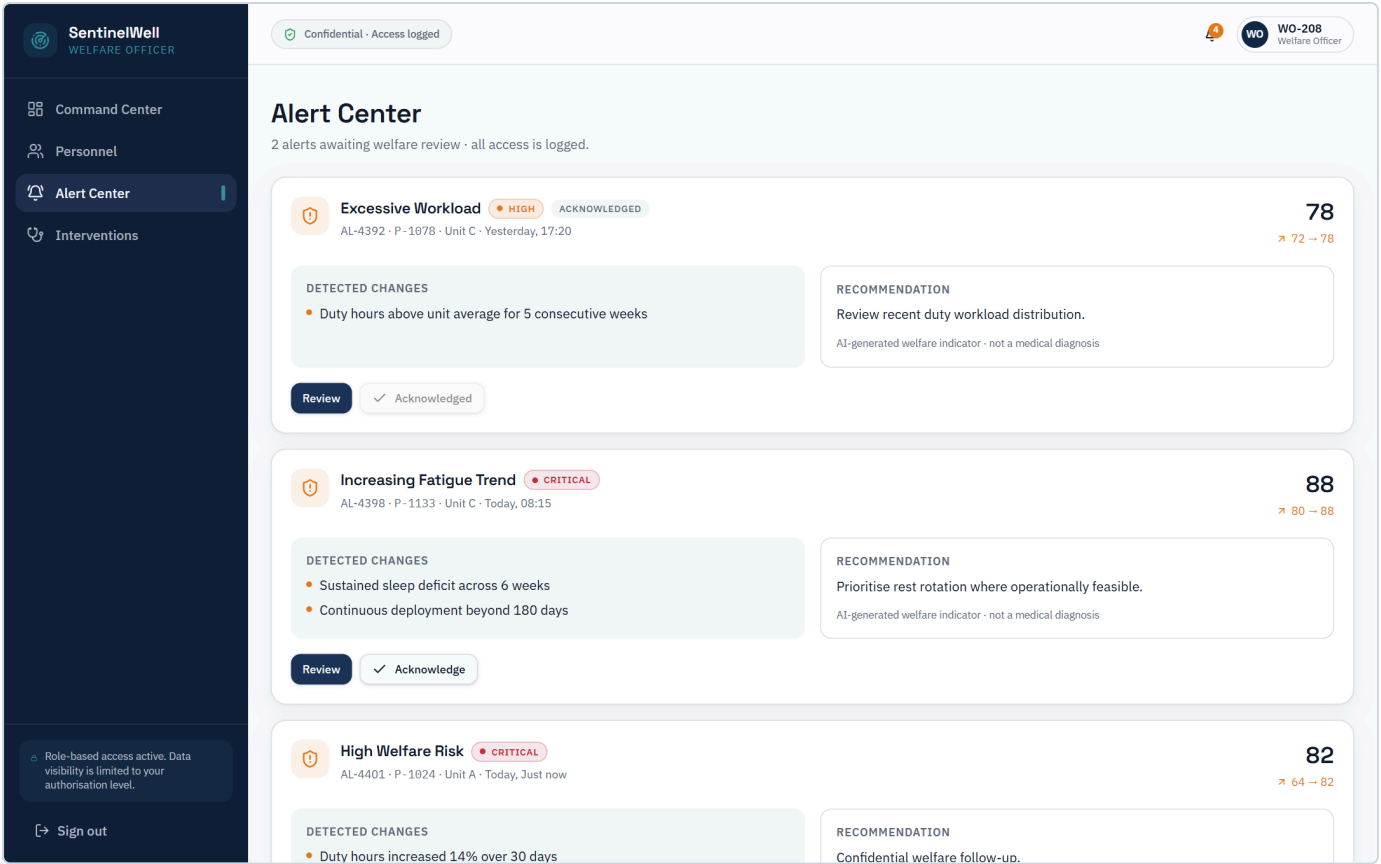


Figure 5: Structured Welfare Interventions Management Table tracking personnel ID, intervention type, assigned officer, date, status, and confidential notes.



Module 4: Commander Battalion Overview & Operational Readiness

Engineered for senior leadership and battalion commanders to assess operational readiness while strictly air-gapping individual personal records:

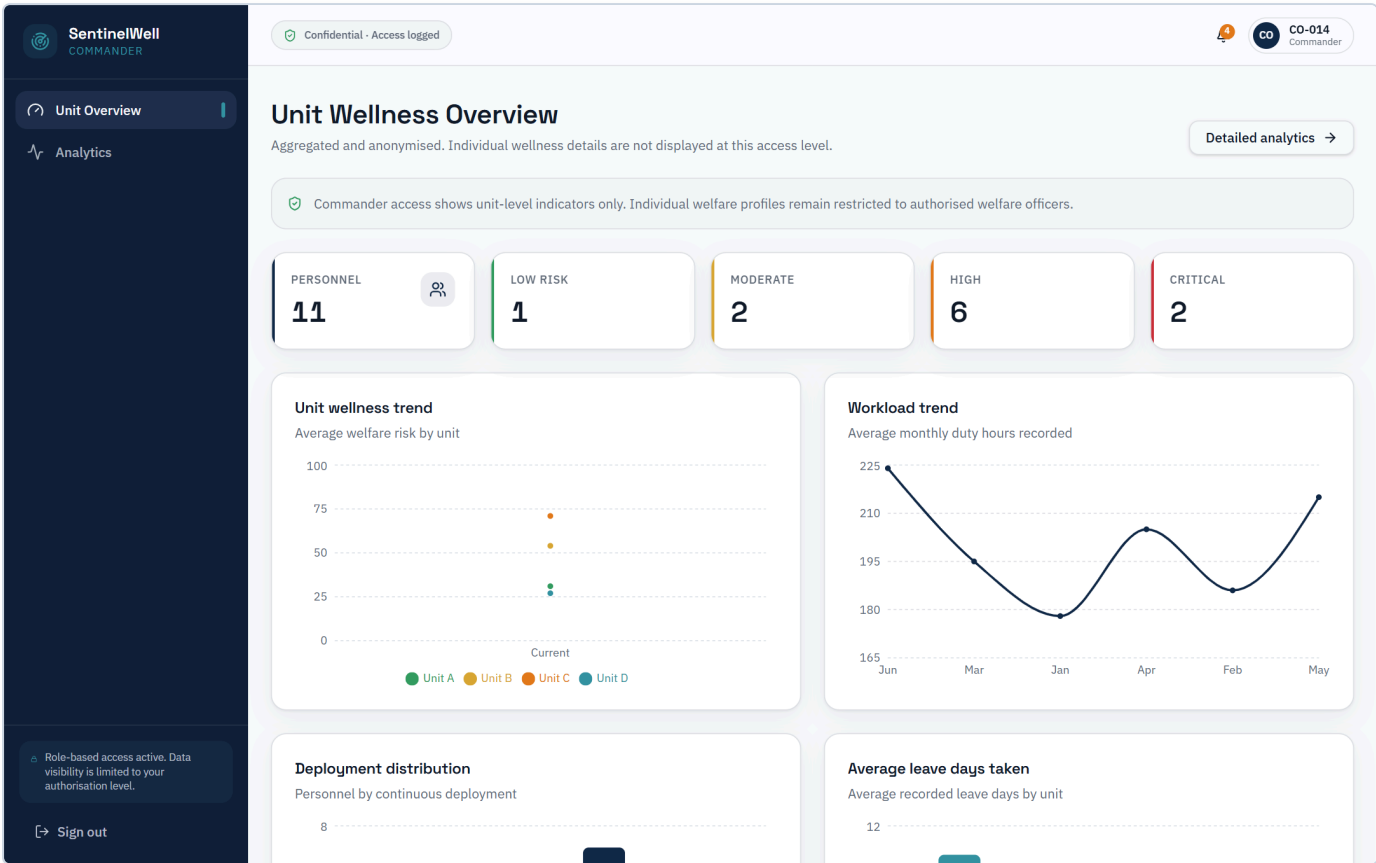


Figure 7: Commander Battalion Overview displaying force readiness, unit wellness index, and comparative monthly duty workload curves.

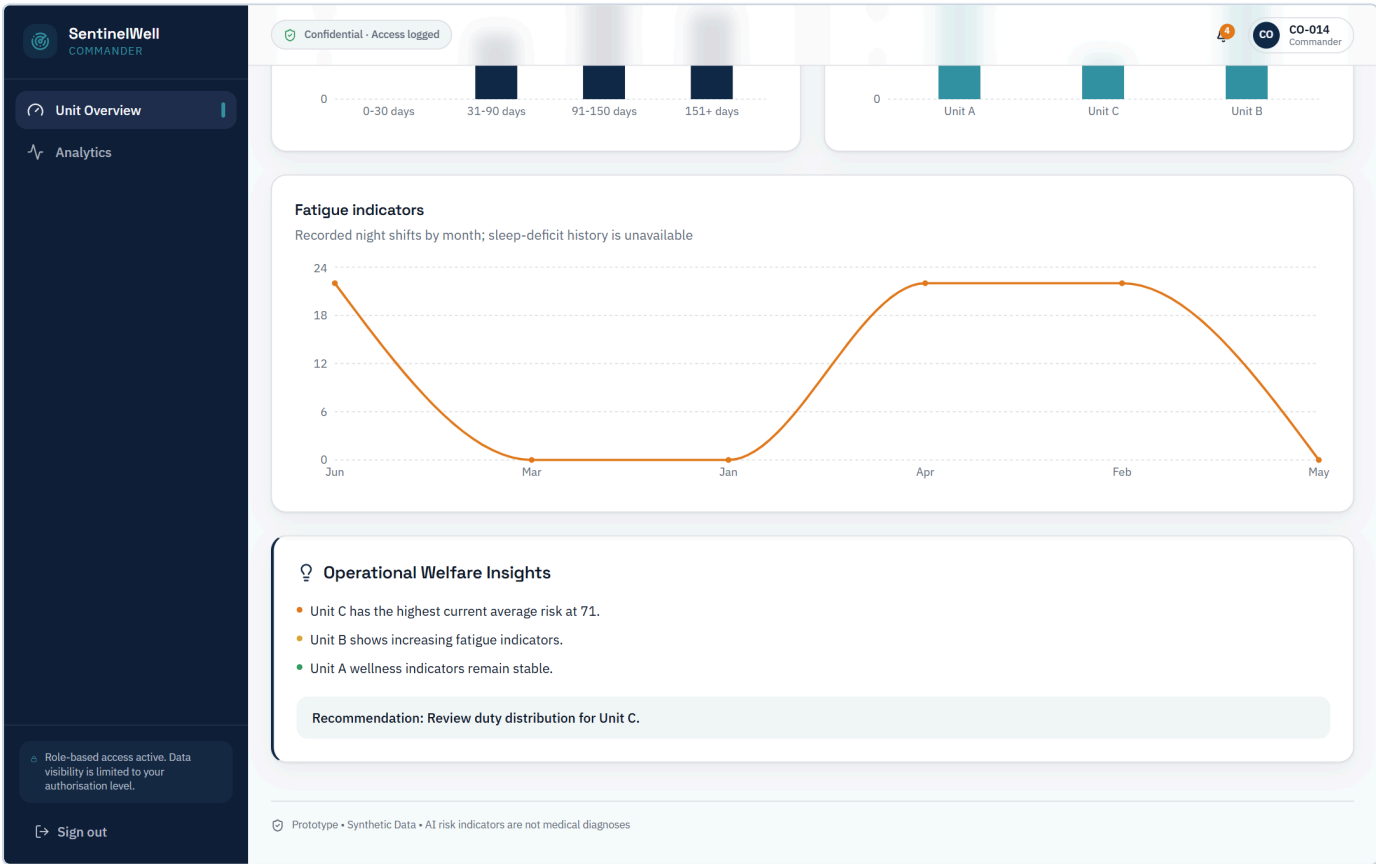
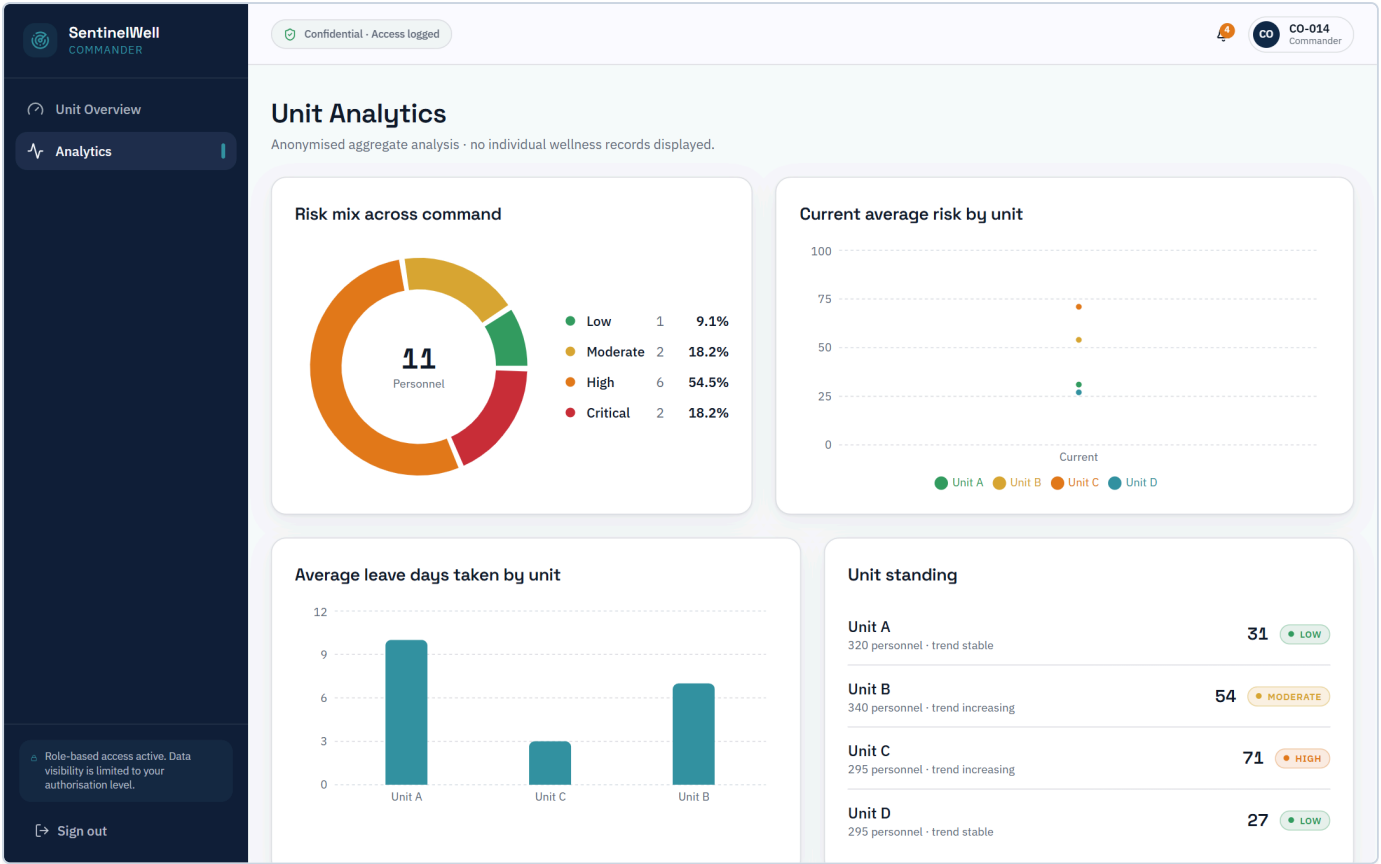


Figure 8: Battalion Fatigue Indicators (recorded night shifts mapped against monthly duty hours) and automated operational recommendations.



Module 5: Personnel Self-Check-In, Self-Trends & Support Portal

A voluntary, confidential mobile-responsive portal empowering jawans and officers with self-monitoring tools:

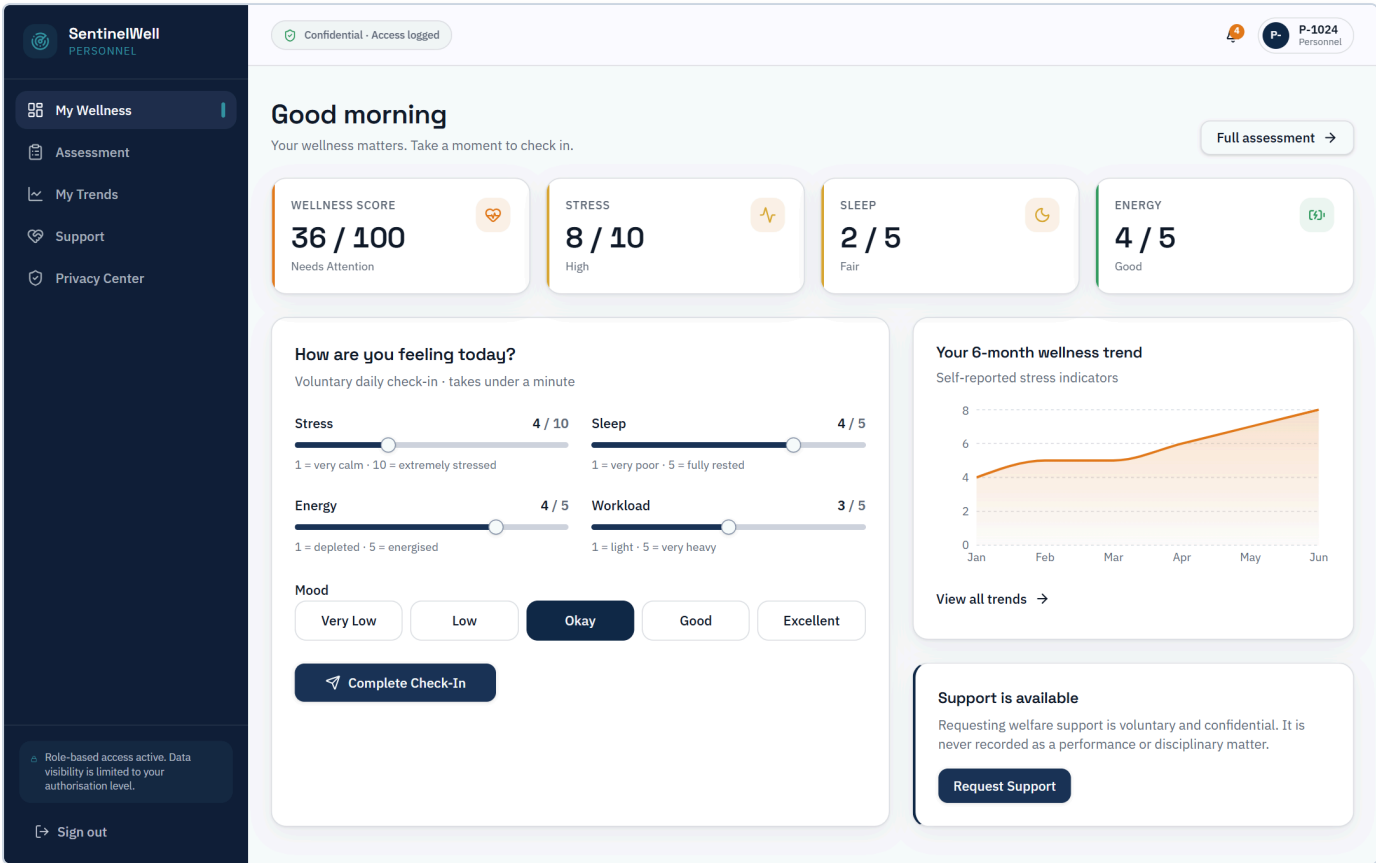


Figure 10: Personnel Wellness Dashboard featuring rapid, sub-minute daily check-in sliders (Stress, Sleep, Energy, Workload).

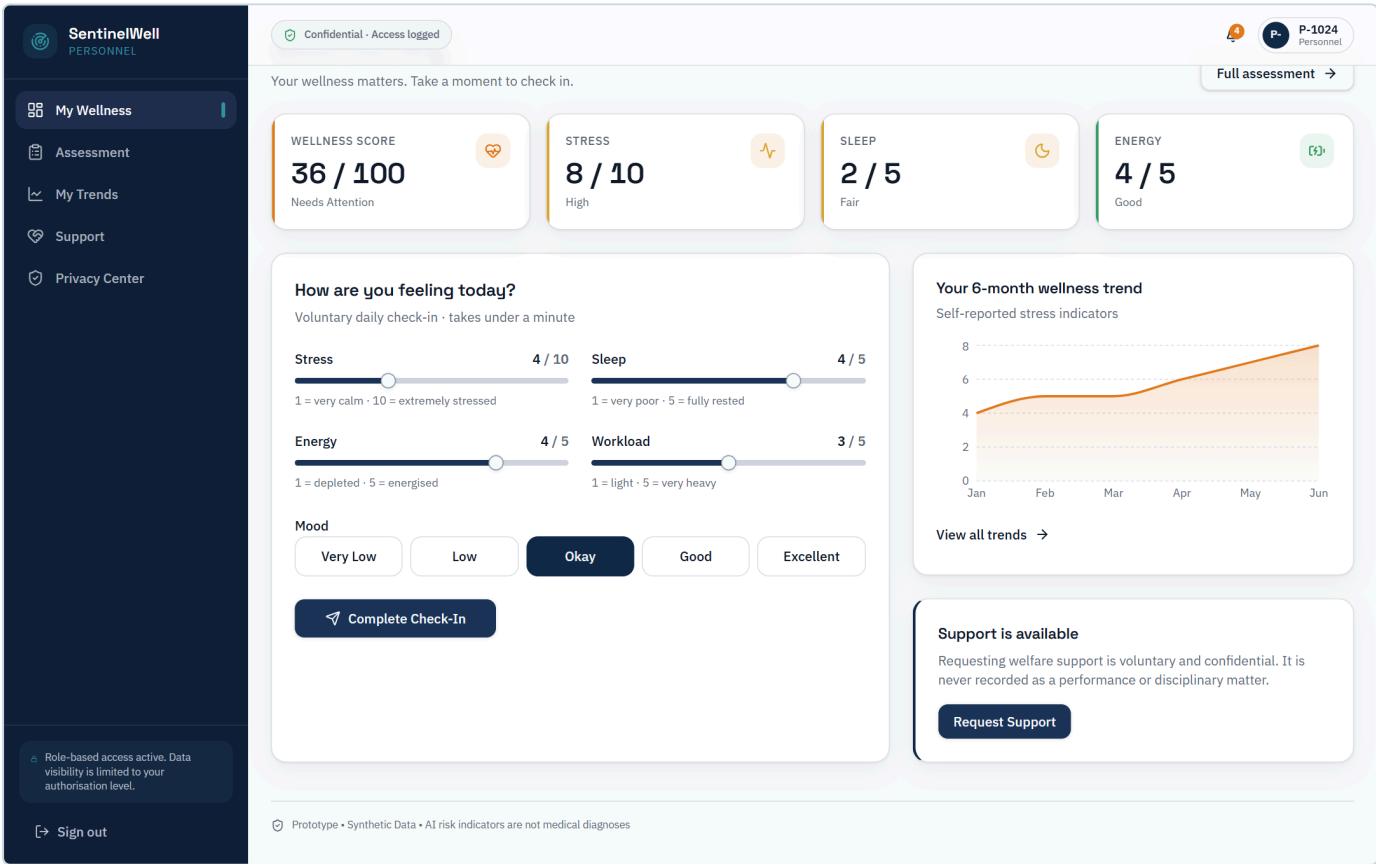


Figure 11: Recent check-in history, historical assessments roster, and direct confidential action cards.

Module 6: Privacy, Governance & Consent Controls

Puts the individual personnel in full control of their data, establishing organizational trust:

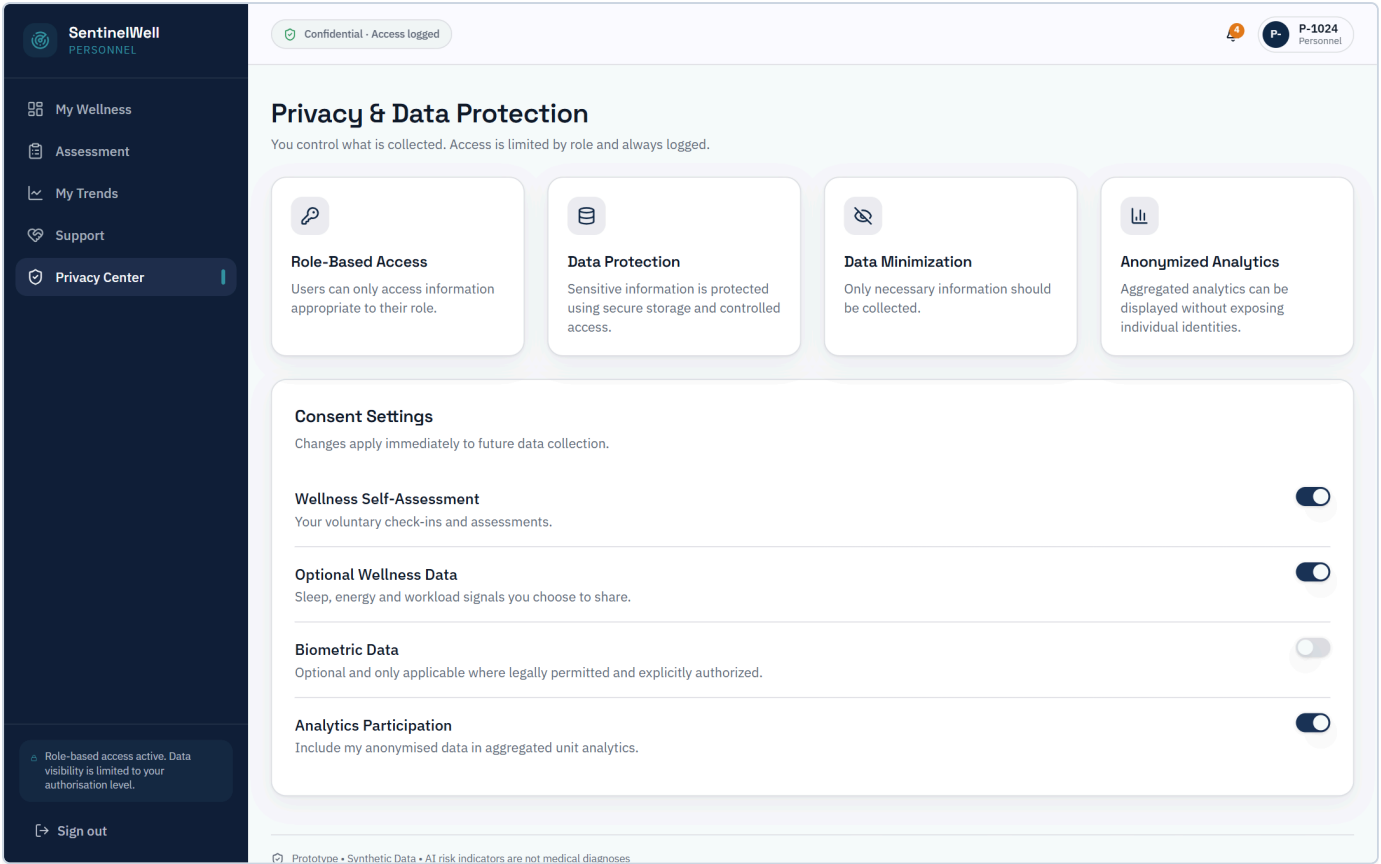


Figure 12: Dedicated Privacy & Data Protection Center with granular toggles for self-assessments, optional wellness data, and zero-biometric default.

Module 7: System Administration, Stress Simulator & Security Audit Trails

Enterprise-level governance tools and live interactive simulation for evaluation and demonstration:

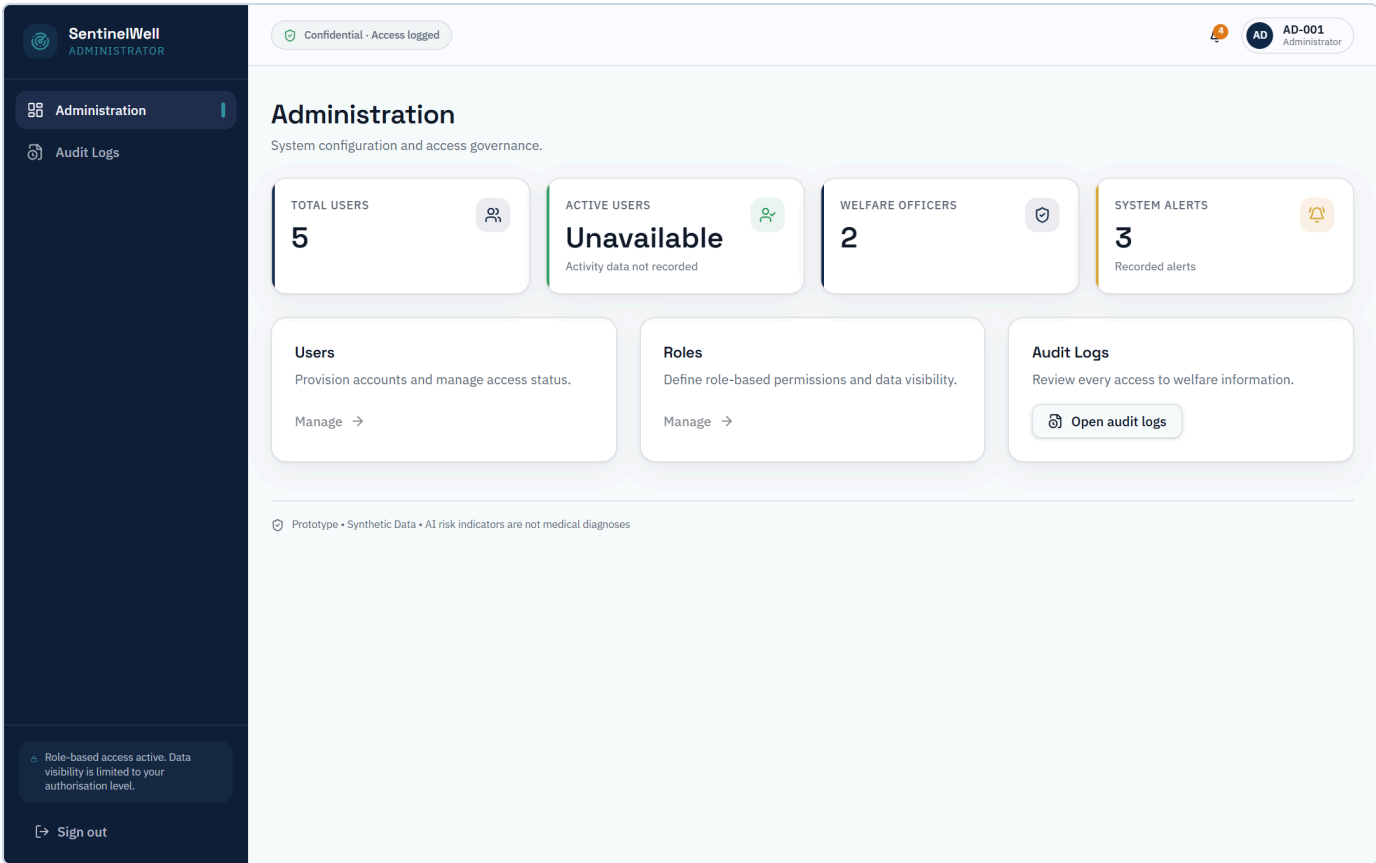


Figure 13: Interactive Stress Simulator demonstrating end-to-end alert propagation upon parameter injection (increasing workload, sleep deficit, night shifts).

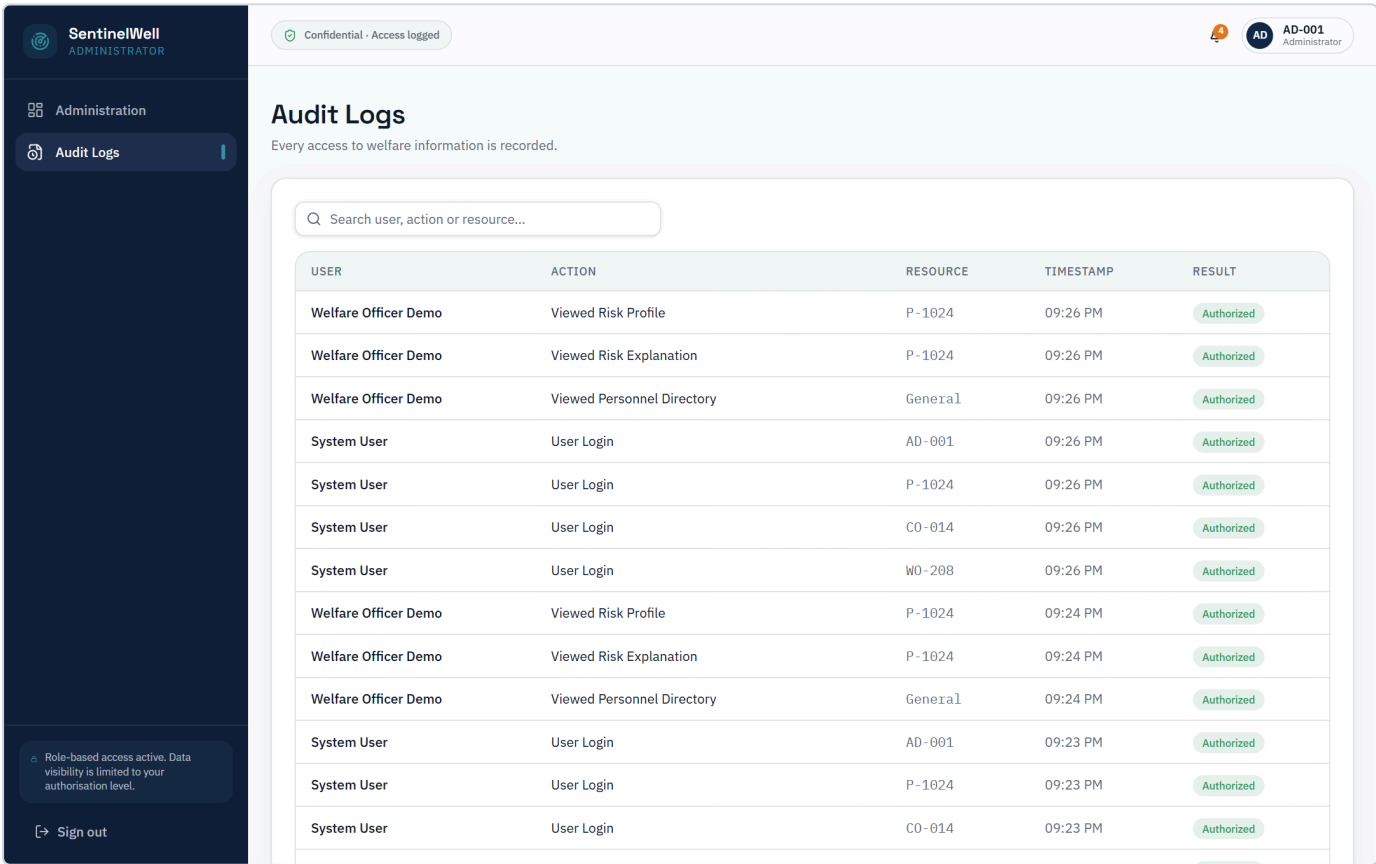


Figure 14: Immutable Security Audit Register displaying timestamped user sessions, resource targets, and authorization results.

6. SECURITY ARCHITECTURE & ROLE-BASED ACCESS CONTROL (RBAC)

Method	API Endpoint	Authorized Roles	Function
POST	/api/auth/login	Public	Signs and returns JWT access + refresh tokens.
GET	/api/personnel	Welfare Officer, Admin	Fetches prioritized welfare triage queue.
GET	/api/personnel/:id	Welfare Officer, Admin	Retrieves individual risk radar & SHAP triggers.
POST	/api/wellness/assessment	Personnel	Records voluntary check-in; invokes ML engine.
GET	/api/analytics/command	Commander, Admin	Fetches anonymized unit readiness curves.
POST	/api/interventions	Welfare Officer	Logs mandatory rest cycle or roster adjustment.

7. PRODUCT & OPERATIONAL ROADMAP

A phased deployment blueprint designed to transition SentinelWell from the SIH prototype to CAPF field formations:

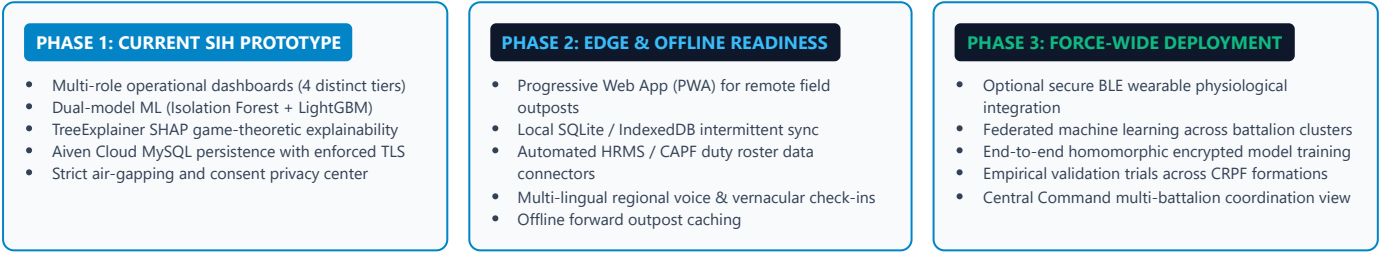


Figure 15: Phased operational roadmap from SIH prototype to national defense deployment.

8. ENGINEERING VERIFICATION TEAM — MANIT BHOPAL

Developed and submitted for the **Smart India Hackathon (SIH) 2026** under Problem Statement ID **26186** (Ministry of Home Affairs / CRPF):

Team Member	Engineering Discipline & Core Responsibilities
Puru Yadav	Team Leader · AI / ML & Predictive Analytics Lead
Nishant Pastor	Frontend Architecture & UI/UX Specialist
Raghav Jhalani	Data Engineering & Pipeline Architecture
HarshVardhan Khare	Model Optimization & Machine Learning Benchmarking
Aarti Misra	Quality Assurance & Defense Domain Research
Sarthak Mittal	Backend Architecture & Cloud Deployment

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